

$a_1, \dots, a_{72}$ of genus 2 in $\mathcal{D}$.

We now define the second kind of surfaces. We first note that every ridge in P is a regular pentagon with angle $\pi/5$. Each of the closed paths just considered crosses 10 pentagonal ridges at their center; it turns out that these are identified in 2 groups of 5, which alternate along the closed path, and give rise to two pentagons in $\mathcal{D}$ whose union is an embedded totally geodesic surface of genus 2. The 72 closed paths give rise in this way to 72 embedded totally geodesic surfaces $A_1, \dots, A_{72}$ in $\mathcal{D}$, again of genus two.

Summing up, there are 72 great circles, and each gives rise to two surfaces a_i and A_i , for $i = 1, \dots, 72$. The surfaces a_i and A_j intersect only if $i = j$, and they do so transversely into two points, that are the centers of two pentagons, with the same sign. Therefore their algebraic intersection is ± 2 , depending on the orientations that we assign to a_i and A_i . Assigning an orientation to a_i is equivalent to assigning an orientation to the corresponding great circle. There does not seem to be a canonical way to orient all the 72 great circles, so we have chosen arbitrarily an orientation for them, and we have oriented the surfaces A_i accordingly so that

$$(4) \quad Q(a_i, A_j) = 2\delta_{ij}.$$

Here Q denotes the intersection form on $H_2(\mathcal{D}, \mathbb{Z})$, and for simplicity we will drop the square brackets from the notation of the homology class of surfaces here and in what follows.

We know from [RT01] that the oriented surfaces A_i forms a basis for $H_2(\mathcal{D}, \mathbb{Q})$, and they generate a subgroup of $H_2(\mathcal{D}, \mathbb{Z}) \cong \mathbb{Z}^{72}$ of index 2^{36} . Any inside-out isometry of $\mathcal{D}$ interchanges the sets $\{a_i\}$ and $\{A_i\}$, however without preserving the index i . Therefore the cycles $\{a_i\}$ also generate a subgroup of $H_2(\mathcal{D}, \mathbb{Z})$ of index 2^{36} , coherently with the fact that the 72×72 diagonal matrix $[Q(a_i, A_j)]$ has determinant 2^{72} . Furthermore, we have

$$(5) \quad Q(a_i, a_j) = \begin{cases} 0 & \text{if the corresponding great circles intersect,} \\ \pm 1 & \text{otherwise.} \end{cases}$$

To see this, we note that if the great circles intersect the two surfaces a_i and a_j can be isotoped to be disjoint, while if they do not intersect the two surfaces a_i and a_j intersect transversely in the center of the 120-cell P . The sign of the intersection is the linking number of the two oriented great circles. For a given a_i , there are exactly 26 classes a_j that have intersection 0 with it: the class a_i itself, and those corresponding to the $10 \times 5/2 = 25$ great circles intersecting it.

Summing up, the 72×72 symmetric intersection matrix $Q_{ij} = Q(a_i, a_j)$ contains only the numbers 0, 1, -1 , with exactly 25 zeroes in every line and column, and it has determinant 2^{72} . We have determined Q explicitly using a computer code.

1.4. Two more classes of surfaces. In order to find more constraints, we will need more surfaces. We keep following [RT01].

The third class of surfaces is constructed as follows: consider for each dodecahedral facet D of P two opposite pentagonal ridges. There are exactly 10 edges of D that do not intersect these two opposite faces; these form an ‘equator’. The identifications of P are such that the edges in D that are identified are precisely the opposite ones (via translation). Because of this, by taking the cone over the center of D , each equator determines a piecewise smooth genus-two surface b_i with a removable self-intersection at the cycle of vertices v . In particular, after a small perturbation b_i becomes a smooth embedded surface of genus 2 in $\mathcal{D}$. There are